JOB SEQUENCING WITH DEADLINE

Problem Statement

- In job sequencing problem, the **objective is to find a sequence of jobs,** which is completed within their deadlines and gives **maximum profit**.

Solution

- Let us consider, **a set of n given jobs** which are associated with **deadlines (d1, d2,...dn) and profit (p1,p2,...pn) is earned, if a job is completed by its deadline.**
- These **jobs need to be ordered** in such a way that there is **maximum profit**.
- . It is also given that **every job takes single unit of time**, so the minimum possible deadline for any job is 1.
- How to maximize total profit if only **one job can be scheduled at a time.**

Here,

- ***We are given a set of jobs.***
- ***Each job has a defined deadline and some profit associated with it.***
- ***The profit of a job is given only when that job is completed within its deadline.***
- ***Only one processor is available for processing all the jobs.***
- ***Processor takes one unit of time to complete a job.***

Approach to Solution-

- A feasible solution would be a subset of jobs where each job of the subset gets completed within its deadline.
- Value of the feasible solution would be the sum of profit of all the jobs contained in the subset.

- An optimal solution of the problem would be a feasible solution which gives the maximum profit.

Greedy Algorithm

- Greedy Algorithm is adopted to determine how the next job is selected for an optimal solution.
- The greedy algorithm described below always gives an optimal solution to the job sequencing problem.

Step-01:

- Sort all the given jobs in decreasing order of their profit.

Step-02:

- Check the value of maximum deadline.
- Draw a Gantt chart where maximum time on Gantt chart is the value of maximum deadline.

Step-03:

- Pick up the jobs one by one.
- Put the job on Gantt chart as far as possible from 0 ensuring that the job gets completed before its deadline.

Example

- Given the jobs, their deadlines and associated profits as shown-

Jobs	J1	J2	J3	J4	J5	J6
Deadlines	5	3	3	2	4	2
Profits	200	180	190	300	120	100

Step-01:

- Sort all the given jobs in decreasing order of their profit-

-

Jobs	J4	J1	J3	J2	J5	J6
Deadlines	2	5	3	3	4	2
Profits	300	200	190	180	120	100

Step-02:

Value of maximum deadline = 5.

So, draw a Gantt chart with maximum time on Gantt chart = 5 units as shown-



Gantt Chart

Now,

- We take each job one by one in the order they appear in Step-01.
- We place the job on Gantt chart as far as possible from 0.

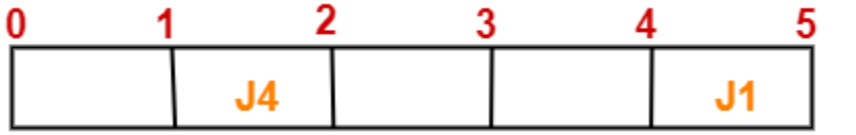
Step-03:

- We take job J4.
- Since its deadline is 2, so we place it in the first empty cell before deadline 2 as-



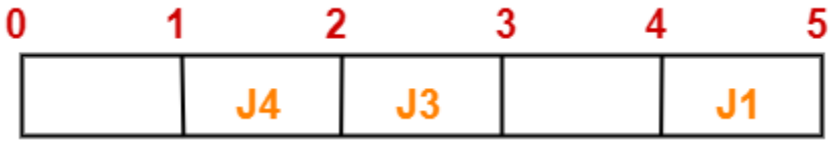
Step-04:

- We take job J1.
- Since its deadline is 5, so we place it in the first empty cell before deadline 5 as-



Step-05:

- We take job J3.
- Since its deadline is 3, so we place it in the first empty cell before deadline 3 as-



Step-06:

- We take job J2.
- Since its deadline is 3, so we place it in the first empty cell before deadline 3.
- Since the second and third cells are already filled, so we place job J2 in the first cell as-



Step-07:

- Now, we take job J5.
- Since its deadline is 4, so we place it in the first empty cell before deadline 4 as-
-



Now,

- The only job left is job J6 whose deadline is 2.
- All the slots before deadline 2 are already occupied.
- Thus, job J6 cannot be completed.

- *The optimal schedule is-*

  {J2, J4, J3, J5, J1}

- This is the required order in which the jobs must be completed in order to obtain the maximum profit.
- *Maximum earned profit*

  = Sum of profit of all the jobs in optimal schedule

  = Profit of job J2 + Profit of job J4 + Profit of job J3 + Profit of job J5 + Profit of job J1

  = 180 + 300 + 190 + 120 + 200=990 units

Example:

Let n = 4,

(P1, P2, P3, P4,) = (100, 10, 15, 27)

(d1 d2 d3 d4) = (2, 1, 2, 1).

Jobs(DECREASING ORDER OF THEIR PROFIT)=J1,J4,J3,J2

1	2
J4	J1

TIMESLOT

FOR I=1 TO 4

K=MIN(DMAX,DEADLINE(1))=MIN(2,2)=2

K=2

WHILE K>=1

IF TIMESLOT[2] IS EMPTY

TIMESLOT[2]= JOB[I]

I=2

K=MIN(DMAX,DEADLINE[2])=MIN(2,1)=1

K=1

WHILE K>=1

IF TIMESLOT[1] IS EMPTY

TIMESLOT[1]=JOB[2]

SOLUTION={J4,J1}

Algorithm-1

- The algorithm constructs an optimal set J of jobs that can be processed by their deadlines.

Algorithm GreedyJob (d, J, n)

// J is a set of jobs that can be completed by their deadlines.

```
{
J: = {1};
for i: = 2 to n do
{
if (all jobs in J U {i} can be completed by their dead lines) then
 J: = J U {i};
}

}
```

Algorithm-2

```
Algorithm GreedyJob (d, J, n)
{
for i =1 to n do

Set k=min (Deadline (i), dmax)//Where Deadline (i) denotes the deadline of
the i^th job
While k>=1 do
If timeslot[k] is empty then

timeslot [k] = job (i)
break
endif
set k=k-1
end while
end for
}
```